

Heap extract-max Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/heap-extract-max

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Abstract

Heap extract-max reduces to unique-maximizer Argmax margin stability: $\gamma > 2\varepsilon$ preserves the winner under $\|\delta\|_\infty \leq \varepsilon$. Lean: `LEAN.FULL`.

1 Theorems

Theorem 1 (Preservation). *Definitional reduction to Argmax margin invariance.*

Theorem 2 (Sharpness). *Definitional reduction to Argmax margin sharpness.*

2 Lean / certificate

`Research.Operators.HeapExtractMax.Margin`.

3 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`); Argmax reduction.